



Brainstorming new predictors for luggage CTR

To kickstart, I dug into L5B15 — let's react to it together

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Starting point: today L5B15 is mostly a *fatigue machine*

Healthy NB model ($AUC \approx 76.6$), but the 23 active predictors lean one way:

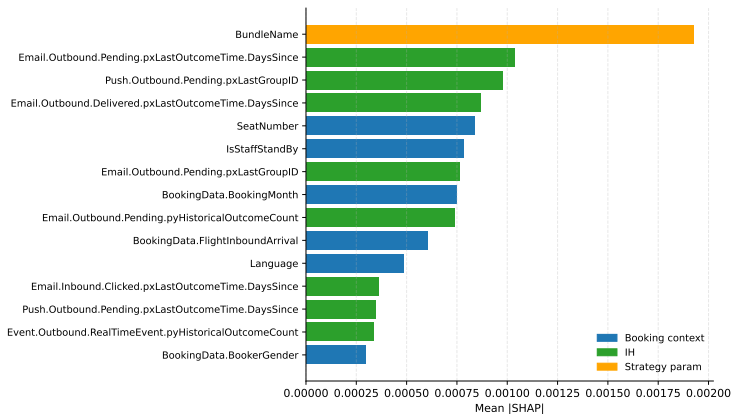
- **11 Interaction History counters** — all Email or Push, mostly Pending
- **11 booking-context fields** — mostly flight metadata
- **1 strategy parameter** (*BundleName*)
- **0 Customer attributes active** (220+ configured, none picked up)

So what drives the score is mostly recency / volume

Almost nothing about the customer or the trip itself. **That's the headroom we want to brainstorm into.**

What the model leans on most today

Top 15 active features by mean |SHAP| on L5B15, coloured by category:



Green = recency / volume of past contact; blue = flight-context; orange = strategy parameter. *No customer-level field is in the list.*

First, what's already inside Pega that the model isn't using?

Pega's `pdstools` has a notebook that ranks *every* field already flowing into the model, not just the 23 NB picked. I ran it on L5B15 ahead of time so we have something to react to. Top candidates currently sitting outside the active set:

Candidate	Imp.	Why it might matter
Booking create date-time	0.109	caveat : likely a timing leak; needs validation
Email-out Delivered, hist. count	0.042	how often Pega <i>tries</i> to email this customer
Booker language	0.025	customer-side, not just flight-context language
Booker role on booking	0.022	primary booker vs. travel companion
Days booking-to-flight	0.020	last-minute vs. planned travel
Email-in Clicked, last group ID	0.020	most recent <i>inbound</i> engagement context
Branded fare bought	0.019	Basic / Plus / Max → willingness to pay

One thing worth flagging

Many *customer-level* fields are 75%+ missing on L5B15. NB drops them as a result — so the gap isn't that they're bad predictors, it's that the data isn't flowing. Re-enabling these is usually the lighter lift.

Then — what could we bring in from outside Pega?

Some guesses to get the

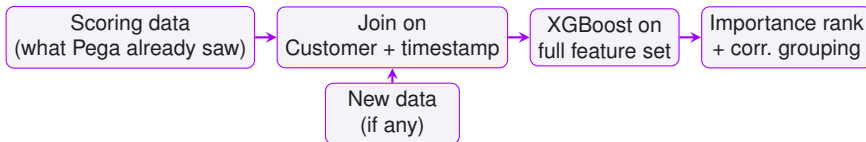
conversation going. Left = why we might expect a lift; right = roughly where the data would have to live.

Why it might help	What we'd add	Where it would come from
Personalisation	Per-product purchase history (luggage, seats, meals, priority)	Already configured as CustomerKPIs.* but mostly missing
Trip purpose	Duration × destination type × children-on-PNR × weekday	Party composition; a destination-type lookup
Route economics	Route type (leisure vs. business); avg luggage attach rate per route	Aggregate attach rate by route, last few months
Price sensitivity	Discount redemptions; ancillary added <i>after</i> booking vs. at booking	Booking-modification log, time-stamped, on PNR
Context	Weather / season at destination on the flight date	A weather table keyed on IATA + date
Loyalty	Tier, months since first booking, distinct destinations flown	Customer master with tier history

Just starting points — add, remove, replace.

How we'd actually test any of this

The same Pega `pdstools` notebook works for both flavours — I can wire each candidate up so we just need to point it at the data:



- No leakage: each row picks the latest snapshot *before* the decision was made
- Verdict per idea: (a) does it rank in the top-20? (b) is it redundant with something already active? (c) does it lift held-out AUC?
- Anything new from outside Pega just needs a Customer/PNR key and a timestamp

Re-running on what's already in Pega is the easier lift; pulling in something new is more work, but that's also where the bigger upside likely sits.

Still open

1. For the inactive fields on slide 4 — are any blocked by an upstream feed (Booker language, Branded fare, Days-to-flight), or is it just a config switch?
2. CustomerKPIs.* is 75 %+ missing. Is Transavia not sourcing it, or is Pega not subscribed? Probably the biggest single unblock.
3. Of the outside-Pega ideas on slide 5 — which already live as tables in the data lake, and which would need to be built?
4. Separately: any appetite to also pilot Pega's **AGB** in shadow mode? That tells us whether the headroom is feature-side or model-side.

Slides 3 & 4 are L5B15 only — happy to run the same view on CLUG / BookingDotCom / Cartrawler if useful.

Over to you

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